

PH102

After Mid Semester Exams

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Chapter-9

EM Waves inside the Matter

EM Waves inside the Matter.

Electromagnetic Wave Propagation in Linear Media

We now consider EM wave propagation inside **linear matter**, but **only** in regions where there are NO **free** charges ($\sigma_{\text{free}} = 0$) and/or **free** currents ($\vec{K}_{\text{free}} = 0$) (i.e. the **linear** medium is an **insulator** / a **non-conductor**).

For this situation, Maxwell's equations (in differential form) become:

$$\nabla \cdot \vec{D} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\vec{D}(r, t) = \epsilon \vec{E}(r, t)$$

$$\vec{H}(r, t) = \frac{1}{\mu} \vec{B}(r, t)$$

$\varepsilon = \text{Electric permittivity of the medium}$

$\mu = \text{Magnetic permittivity of the medium}$

$\varepsilon = \varepsilon_0(1 + \chi_e), \quad \chi_e = \text{Electric Susceptibility of the medium}$

$\mu = \mu_0(1 + \chi_m), \quad \chi_m = \text{Magnetic permeability of the medium}$

$\varepsilon_0 = \text{Electrical permittivity of free space} = 8.85 \times 10^{-12} \text{ Farads/m}$

$\mu_0 = \text{Magnetic permittivity of free space} = 4\pi \times 10^{-7} \text{ Henrys/m}$

Thus, Maxwell's equations for the $\vec{E}$ and $\vec{B}$ fields inside the linear, homogeneous and isotropic non-conducting medium become:

$$\nabla \cdot \vec{E}(r, t) = 0$$

$$\nabla \times \vec{E}(r, t) = -\frac{\partial \vec{B}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{B}(r, t) = 0$$

$$\nabla \times \vec{B}(r, t) = \mu\varepsilon \frac{\partial \vec{E}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{E}(r, t) = 0$$

$$\nabla \times \vec{E}(r, t) = -\frac{\partial \vec{B}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{B}(r, t) = 0$$

$$\nabla \times \vec{B}(r, t) = \mu\epsilon \frac{\partial \vec{E}(r, t)}{\partial t}$$

Note that the above four relations are (almost) identical to those for *EM* waves in free space.

We simply replace the macroscopic *EM* parameters associated with the vacuum with $\{\epsilon_0, \mu_0\}$ those associated with the linear, homogeneous and isotropic medium $\{\epsilon, \mu\}$.

In free space/vacuum, the speed of propagation of *EM* waves is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \frac{\text{m}}{\text{s}}$$

$\vec{E}$ and $\vec{B}$ fields in vacuum obey the wave equation

$$\nabla^2 \vec{E} = \varepsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \varepsilon_0 \mu_0 \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

In a linear/homogeneous/isotropic medium, the speed of propagation of *EM* waves is:

$$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

The **E** and **B** fields in the medium obey the following wave equation:

$$\nabla^2 \vec{E} = \varepsilon \mu \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \varepsilon \mu \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

For linear/homogeneous/isotropic media

$$\varepsilon = K_e \varepsilon_0 = (1 + \chi_e) \varepsilon_0 \quad K_e = \frac{\varepsilon}{\varepsilon_0}$$

Relative electric permittivity

$$\mu = K_m \mu_0 = (1 + \chi_m) \mu_0 \quad K_m = \frac{\mu}{\mu_0}$$

Relative magnetic permeability

$$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

$$v = \frac{1}{\sqrt{K_e \varepsilon_0 K_m \mu_0}} = \frac{1}{\sqrt{K_e K_m}} \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \frac{c}{\sqrt{K_e K_m}}$$

Now if:

$$K_e = \frac{\varepsilon}{\varepsilon_0} = (1 + \chi_e) \geq 1 \quad K_m = \frac{\mu}{\mu_0} = (1 + \chi_m) \geq 1$$

$$K_e K_m \geq 1$$

True for a wide variety of common/everyday materials – gases, liquids & solids

$$\sqrt{K_e K_m} \geq 1$$



$$\frac{1}{\sqrt{K_e K_m}} \leq 1$$

$$v = \frac{c}{\sqrt{K_e K_m}} \leq c$$

We can now define the index of refraction {*n.b.* a dimensionless quantity} of the linear/homogeneous/isotropic medium as:

$$n = \sqrt{K_e K_m} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}}$$

Thus, for linear/homogeneous/isotropic media:

$$v = \frac{c}{n} \leq c$$

$$n \geq 1 \text{ (usually)}$$

Usually ε and μ are not constant but they are frequency dependent

$$\varepsilon = \varepsilon(\omega)$$

$$\mu = \mu(\omega)$$

$$\omega = 2\pi f$$

$$K_e = K_e(\omega) = \frac{\varepsilon(\omega)}{\varepsilon_0} = (1 + \chi_e(\omega))$$

$$K_m = K_m(\omega) = \frac{\mu(\omega)}{\mu_0} = (1 + \chi_m(\omega))$$

$$n = n(\omega) = \sqrt{K_e(\omega)K_m(\omega)} = \sqrt{\frac{\varepsilon(\omega)\mu(\omega)}{\varepsilon_0\mu_0}} = \sqrt{(1 + \chi_e(\omega))(1 + \chi_m(\omega))}$$

$$v(\omega) = \frac{c}{n(\omega)}$$

For now, we will ignore frequency-dependent effects, for simplicity, *i.e.*

$$v(\omega) = \frac{c}{n} = \textit{Constant}$$

$$n = \sqrt{(1 + \chi_e)(1 + \chi_m)} = \textit{Constant}$$

Now for many (but not all) linear/homogeneous/isotropic materials

$$\mu = (1 + \chi_m)\mu_0 \approx \mu_0$$

This is true for many paramagnetic and diamagnetic-type materials since $\chi_m \sim 10^{-8} \sim 0$

$$K_m = \frac{\mu}{\mu_0} = 1$$

$$n = \sqrt{K_e K_m} = \sqrt{K_e}$$

$$v = \frac{c}{n} = \frac{c}{\sqrt{K_e}}$$

The instantaneous EM energy density associated with a linear/homogeneous/isotropic material:

$$u_{em} = (u_e + u_m)$$

$$u_{em} = (u_e + u_m)$$

$$u_{em} = \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H})$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{B^2}{\mu} \right)$$

Joules/m³

$$\vec{D} = \epsilon \vec{E} \quad \vec{H} = \frac{1}{\mu} \vec{B}$$

But

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E} \quad \& \quad v^2 = \frac{1}{\epsilon \mu}$$

$$v^2 = \frac{1}{\mu\epsilon}$$

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E} \quad \& \quad v^2 = \frac{1}{\epsilon\mu}$$

Hence

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{E^2}{\mu v^2} \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{B^2}{\mu} \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{\mu\epsilon E^2}{\mu} \right)$$

$$u_{em} = \frac{1}{2} (\epsilon E^2 + \epsilon E^2)$$

$$u_{em} = \epsilon E^2$$

Hence for linear media

$$u_{em} = \frac{1}{2} (\epsilon E^2 + \mu H^2)$$

$$u_e = u_m$$

The instantaneous Poynting's vector associated with a linear/homogeneous/isotropic material:

$$\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu} (\vec{E} \times \vec{B})$$

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E}$$

$$\vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B}) = \frac{1}{\mu v} E^2 \hat{k}$$

$$v^2 = \frac{1}{\mu \epsilon}$$

$$\vec{S} = \frac{v}{\mu v^2} E^2 \hat{k}$$

$$\vec{S} = \frac{v}{\mu v^2} E^2 \hat{k}$$

$$v^2 = \frac{1}{\mu \epsilon}$$

$$\vec{S} = v \epsilon E^2 \hat{k}$$

$$u_{em} = \epsilon E^2$$

$$\vec{S} = v u_{em} \hat{k}$$

Lets define the Vector impedance associated with a linear/homogeneous/isotropic material is given by the relation:

$$\vec{Z} = \vec{E} \times \vec{H}^{-1}$$

$$\vec{Z} = \frac{(\vec{E} \times \vec{H})}{|\vec{H}|^2}$$

$$\vec{Z} = \frac{(\vec{E} \times \vec{H})}{|\vec{H}|^2}$$

$$\vec{S} = v u_{em} \hat{k}$$

$$u_{em} = \varepsilon E^2$$

$$\vec{S} = v \varepsilon E^2 \hat{k}$$

$$\vec{Z} = \mu v \hat{k}$$

$$\vec{Z} = \sqrt{\frac{\mu}{\varepsilon}} \hat{k}$$

Ohms

For monochromatic (*i.e.* sinusoidal, single frequency) plane *EM* waves propagating in a linear/homogeneous/isotropic medium, **E** and **B** satisfy/obey the wave equation:

$$\nabla^2 \vec{E} = \varepsilon \mu \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

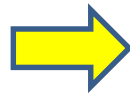
$$\nabla^2 \vec{B} = \varepsilon \mu \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

E and B -field solutions of wave eqn. for linearly polarized plane EM wave with Polarization $\hat{n} \perp \hat{k}$ propagating in this linear/homogeneous/isotropic medium are of the form:

$$\mathbf{E}(\mathbf{r}, t) = E_0 \cos(\vec{k} \cdot \vec{r} - \omega t + \delta) \hat{n}$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} E_0 \cos(\vec{k} \cdot \vec{r} - \omega t + \delta) (\hat{k} \times \hat{n})$$

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E}$$



$$|\vec{B}| = \frac{1}{v} |\vec{E}|$$



$$B_0 = \frac{1}{v} E_0$$

And

$$v = f\lambda = \frac{\omega}{k}$$

$$\omega = 2\pi f$$

$$k = 2\pi/\lambda$$

The intensity of an EM wave propagating in a linear/homogeneous/isotropic medium is:

$$\mathbf{I}(\mathbf{r}, t) = \langle |\vec{S}| \rangle = v \langle u_{em} \rangle$$

$$u_{em} = \epsilon E^2$$

$$\mathbf{I}(\mathbf{r}, t) = \frac{1}{2} v \epsilon E_0^2 = \frac{1}{2} \left(\frac{c}{n} \right) \epsilon E_0^2$$

$$E_{0-rms} = \frac{1}{\sqrt{2}} E_0$$

$$\mathbf{I}(\mathbf{r}, t) = \left(\frac{c}{n} \right) \epsilon E_{0-rms}^2$$

Watts/m²

RMS intensity of the EM wave

$$\mathbf{I}_{\text{rms}}(\mathbf{r}, t) = \langle |\vec{S}_{\text{rms}}| \rangle = v \langle |\vec{u}_{\text{em-rms}}| \rangle$$

$$\mathbf{I}_{\text{rms}}(\mathbf{r}, t) = \frac{1}{2} v \epsilon E_{0-\text{rms}}^2 = \frac{1}{2} \left(\frac{c}{n} \right) \epsilon E_{0-\text{rms}}^2$$

Rms-Watts/m²

$$u_{\text{em}} = \epsilon E^2$$

$$\mathbf{I}_{\text{rms}}(\mathbf{r}, t) = \frac{1}{2} \mathbf{I}(\mathbf{r}, t)$$

$$\langle |\vec{S}_{\text{rms}}| \rangle = \frac{1}{2} \langle |\vec{S}| \rangle$$

$$\langle u_{\text{em-rms}} \rangle = \frac{1}{2} \langle u_{\text{em}} \rangle$$

Example: What happens when an *EM* wave passes from one linear/homogeneous/isotropic medium into another (e.g. vacuum $\rightarrow$ gas; air $\rightarrow$ water; water $\rightarrow$ oil; glass $\rightarrow$ plastic; *etc...*?)

Solution: As we saw in the case of mechanical transverse traveling waves propagating on the taught string which had two different mass-per-unit-lengths μ_1 and μ_2 , we anticipate that *EM* wave ***reflection*** and wave ***transmission*** phenomena will also occur at the interface/boundary between two different linear/homogeneous/isotropic media.

However, in the *EM* wave situation, what actually happens at the boundary/interface between two linear/homogeneous/isotropic media depends on the electro-dynamic versions of the boundary conditions on the **E** and **B**-fields at that interface

Boundary Condition 1: The normal component of $\mathbf{D}$ is continuous across the interface @ $z = 0$ (true only when there are no free surface charges present @ the interface $\sigma_{\text{free}} = 0$).

$$D_1^\perp = D_2^\perp \quad \text{At interface}$$

$$\epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp$$

Boundary Condition 2: The TANGENTIAL component of $\mathbf{E}$ is continuous across interface @ $z = 0$:

$$E_1^\parallel = E_2^\parallel \quad \text{At interface}$$

Boundary Condition 3: The NORMAL component of $\mathbf{B}$ is continuous across the interface @ $z = 0$:

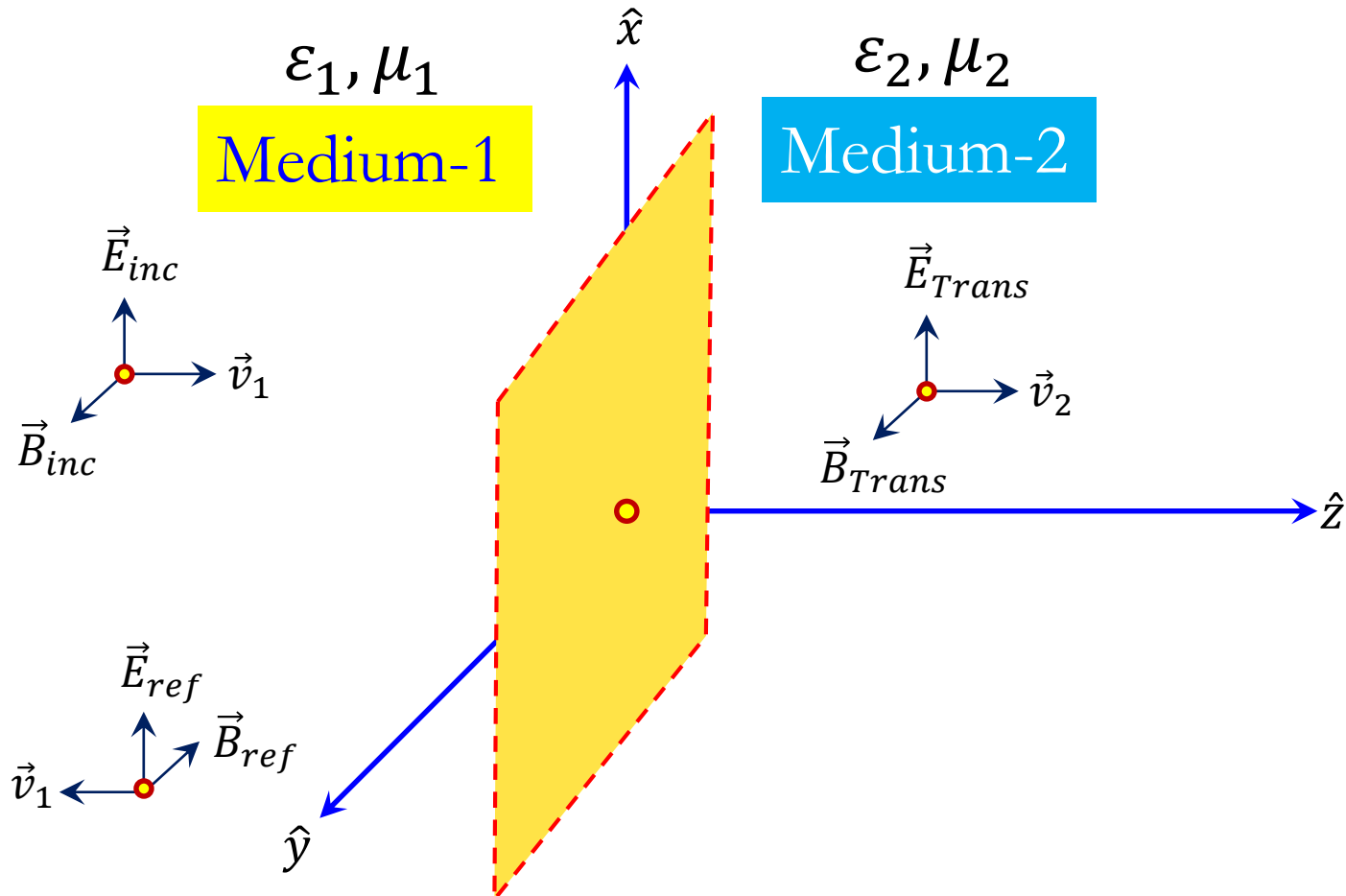
$$B_1^\perp = B_2^\perp \quad \text{At interface}$$

Boundary Condition 4: The TANGENTIAL component of $\mathbf{H}$ is continuous across the interface @ $z = 0$

$$H_1^\parallel = H_2^\parallel \quad \text{At interface}$$

$$\frac{1}{\mu_1} B_1^\parallel = \frac{1}{\mu_2} B_2^\parallel$$

Reflection & Transmission of Linear Polarized Plane EM Waves at Normal Incidence at a Boundary Between Two Linear/Homogeneous /Isotropic Media



Incident EM plane wave (in medium 1):

Propagates in the $+\hat{z}$ -direction with polarization $\hat{n}_{inc} = +\hat{x}$

$$\vec{\tilde{E}}_{inc} = \tilde{E}_{0-inc} e^{i(k_1 z - \omega t)} \hat{x} \quad \text{with} \quad k_{inc} = |\vec{k}_{inc}| = |\vec{k}_1| = \frac{2\pi}{\lambda_1} = \frac{\omega}{v_1}$$

$$\vec{\tilde{B}}_{inc} = \frac{1}{v_1} \hat{k}_{inc} \times \vec{\tilde{E}}_{inc} = \frac{1}{v_1} \tilde{E}_{0-inc} e^{i(k_1 z - \omega t)} \hat{y}$$

Reflected EM plane wave (in medium 1):

Propagates in the $-\hat{z}$ -direction with polarization $\hat{n}_{ref} = +\hat{x}$

$$\vec{\tilde{E}}_{ref} = \tilde{E}_{0-ref} e^{i(-k_1 z - \omega t)} \hat{x} \quad \text{with} \quad k_{ref} = |\vec{k}_{ref}| = |\vec{k}_1| = \frac{2\pi}{\lambda_1} = \frac{\omega}{v_1}$$

$$\vec{\tilde{B}}_{ref} = \frac{1}{v_1} \hat{k}_{ref} \times \vec{\tilde{E}}_{ref} = -\frac{1}{v_1} \tilde{E}_{0-ref} e^{i(-k_1 z - \omega t)} \hat{y}$$

Transmitted EM plane wave (in medium 2):

Propagates in the $+\hat{z}$ -direction with polarization $\hat{n}_{tra} = +\hat{x}$

$$\vec{\tilde{E}}_{tra} = \tilde{E}_{0-tra} e^{i(k_2 z - \omega t)} \hat{x} \quad \text{with} \quad k_{tra} = |\vec{k}_{tra}| = |\vec{k}_2| = \frac{2\pi}{\lambda_2} = \frac{\omega}{v_2}$$

$$\vec{\tilde{B}}_{tra} = \frac{1}{v_2} \hat{k}_{tra} \times \vec{\tilde{E}}_{tra} = \frac{1}{v_1} \tilde{E}_{0-tra} e^{i(k_2 z - \omega t)} \hat{y}$$

Note that {**here**, in **this** situation} the E-field/**polarization** vectors are all oriented in the **same** direction, i.e.

$$\hat{n}_{inc} = \hat{n}_{ref} = \hat{n}_{tra} = +\hat{x}$$

Or equivalently

$$\vec{\tilde{E}}_{inc} \parallel \vec{\tilde{E}}_{ref} \parallel \vec{\tilde{E}}_{tra}$$

At the interface/boundary between the two linear/homogeneous/isotropic media, *i.e.* at $z = 0$ {in the x - y plane} the boundary conditions (1) to (4) **must** be satisfied for the total $\mathbf{E}$ and $\mathbf{B}$ -fields immediately present on either side of the interface between the two media:

Boundary Condition 1: The normal component of $\mathbf{D}$ is continuous across the interface @ $z = 0$ (true only when there are no free surface charges present @ the interface $\sigma_{\text{free}} = 0$).

$$D_1^\perp = D_2^\perp \quad \text{At interface } (z=0, t)$$

$$\epsilon_1 E_{1-\text{Tot}}^\perp = \epsilon_2 E_{2-\text{Tot}}^\perp$$

Boundary Condition 2: The TANGENTIAL component of $\mathbf{E}$ is continuous across interface @ $z = 0$:

$$E_{1-\text{Tot}}^\parallel = E_{2-\text{Tot}}^\parallel \quad \text{At interface } (z=0, t)$$

Boundary Condition 3: The NORMAL component of **B** is continuous across the interface (@ $z = 0$):

$$B_{1-\text{Tot}}^{\perp} = B_{2-\text{Tot}}^{\perp} \quad \text{At interface (z=0,t)}$$

Boundary Condition 4: The TANGENTIAL component of **H** is continuous across the interface (@ $z = 0$)

$$H_{1-\text{Tot}}^{\parallel} = H_{2-\text{Tot}}^{\parallel} \quad \text{At interface (z=0,t)}$$

$$\frac{1}{\mu_1} B_{1-\text{Tot}}^{\parallel} = \frac{1}{\mu_2} B_{2-\text{Tot}}^{\parallel}$$

Normal incidence

For plane EM waves at **normal** incidence on the boundary at $z = 0$ lying in the x - y plane, note that **no** components of E or B (incident, reflected or transmitted waves) are allowed to be along the $\hat{z}$ propagation direction(s) because of the E and B -field **transversally** requirement(s) on the propagation of EM waves {arising from the constraints imposed by Maxwell's equations}.

Thus, because of this, we see that **Boundary Conditions** (1) and (3) impose **no** restrictions {here} on such EM waves.

The only restrictions on plane EM waves propagating with normal incidence on the boundary at $z = 0$ {lying in the x-y plane} are imposed by **Boundary Conditions** (2) and (4).

In medium (1) (*i.e.* $z \leq 0$):

$$E_{1-Tot}^{\parallel} = \vec{\tilde{E}}_{1-inc} + \vec{\tilde{E}}_{1-ref}$$
$$\frac{1}{\mu_1} B_{1-Tot}^{\parallel} = \frac{1}{\mu_1} \vec{\tilde{B}}_{inc} + \frac{1}{\mu_1} \vec{\tilde{B}}_{ref}$$

In medium (2) (*i.e.* $z \geq 0$):

$$E_{2-Tot}^{\parallel} = \vec{\tilde{E}}_{tra}$$
$$\frac{1}{\mu_2} B_{2-Tot}^{\parallel} = \frac{1}{\mu_2} \vec{\tilde{B}}_{tra}$$

These relations also hold/are valid on the boundary, *i.e.* @ $z = 0$.

Then the boundary condition (2) (Tangential $\mathbf{E}$ is continuous on the boundary @ $z = 0$) requires that

$$\overrightarrow{\tilde{E}_{1-tot}} \Big|_{z=0}^{\parallel} = \overrightarrow{\tilde{E}_{2-tot}} \Big|_{z=0}^{\parallel}$$

$$\vec{\tilde{E}}_{inc} + \vec{\tilde{E}}_{ref} = \vec{\tilde{E}}_{tra} \Big|_{z=0}^{\parallel}$$

And boundary condition (4) (Tangential $\mathbf{H}$ is continuous on the boundary @ $z = 0$) requires that

$$\frac{1}{\mu_1} \overrightarrow{\tilde{B}_{1-tot}} \Big|_{z=0}^{\parallel} = \frac{1}{\mu_2} \overrightarrow{\tilde{B}_{2-tot}} \Big|_{z=0}^{\parallel}$$

$$\frac{1}{\mu_1} \vec{\tilde{B}}_{inc} + \frac{1}{\mu_1} \vec{\tilde{B}}_{ref} = \frac{1}{\mu_2} \vec{\tilde{B}}_{tra} \Big|_{z=0}^{\parallel}$$

Inserting the explicit expressions for the complex **E** and **B** fields into the above boundary condition conditions, these equations become:

$$\tilde{E}_{0-inc}e^{i(-\omega t)} + \tilde{E}_{0-ref}e^{i(-\omega t)} = \tilde{E}_{0-tra}e^{i(-\omega t)}$$

$$\tilde{E}_{0-inc} + \tilde{E}_{0-ref} = \tilde{E}_{0-tra}$$

$$\frac{1}{\mu_1 v_1} \tilde{E}_{0-inc}e^{i(-\omega t)} - \frac{1}{\mu_1 v_1} \tilde{E}_{0-ref}e^{i(-\omega t)} = \frac{1}{\mu_2 v_2} \tilde{E}_{0-tra}e^{i(-\omega t)}$$

$$\frac{1}{\mu_1 v_1} \tilde{E}_{0-inc} - \frac{1}{\mu_1 v_1} \tilde{E}_{0-ref} = \frac{1}{\mu_2 v_2} \tilde{E}_{0-tra}$$

Solving these equations simultaneously for $\{\tilde{E}_{0-ref}$ and $\tilde{E}_{0-tra}$ $\}$ in terms of/scaled to $\tilde{E}_{0-inc}$.

First (for convenience) let us define:

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} \quad \Rightarrow \quad \begin{matrix} Z_1 = \mu_1 v_1 \\ Z_2 = \mu_2 v_2 \end{matrix} \quad \Rightarrow \quad \beta = \frac{Z_1}{Z_2}$$

Then Boundary Condition (4); (Tangential H continuous @ $z = 0$) relation becomes:

$$\frac{1}{\mu_1 v_1} \tilde{E}_{0-inc} - \frac{1}{\mu_1 v_1} \tilde{E}_{0-ref} = \frac{1}{\mu_2 v_2} \tilde{E}_{0-tra}$$



$$\tilde{E}_{0-inc} - \tilde{E}_{0-ref} = \beta \tilde{E}_{0-tra}$$

$$\tilde{E}_{0-inc} - \tilde{E}_{0-ref} = \beta \tilde{E}_{0-tra}$$

$$\tilde{E}_{0-inc} + \tilde{E}_{0-ref} = \tilde{E}_{0-tra}$$

On solving these two equation we get

$$2\tilde{E}_{0-inc} = (1 + \beta)\tilde{E}_{0-tra}$$



$$\tilde{E}_{0-tra} = \frac{2}{1 + \beta} \tilde{E}_{0-inc}$$

$$2\tilde{E}_{0-ref} = (1 - \beta)\tilde{E}_{0-tra}$$



$$\tilde{E}_{0-ref} = \frac{1 - \beta}{2} \tilde{E}_{0-tra}$$

$$\tilde{E}_{0-ref} = \frac{1 - \beta}{1 + \beta} \tilde{E}_{0-inc}$$

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{Z_1}{Z_2}$$

$$v_1 = \frac{c}{n_1}$$

$$v_2 = \frac{c}{n_2}$$

$$n_1 = \sqrt{\frac{\epsilon_1 \mu_1}{\epsilon_0 \mu_0}}$$

$$n_2 = \sqrt{\frac{\epsilon_2 \mu_2}{\epsilon_0 \mu_0}}$$

$$\beta = \frac{\mu_1 (c/n_1)}{\mu_2 (c/n_2)} = \frac{\mu_1 n_2}{\mu_2 n_1} = \frac{\mu_1 \sqrt{\frac{\epsilon_2 \mu_2}{\epsilon_0 \mu_0}}}{\mu_2 \sqrt{\frac{\epsilon_1 \mu_1}{\epsilon_0 \mu_0}}} = \sqrt{\frac{\epsilon_2 \mu_1}{\epsilon_1 \mu_2}}$$

Now if the two media are **both** paramagnetic and/or diamagnetic, such that

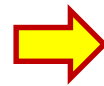
$$\mu_1 = (1 + \chi_{m1})\mu_0 \approx \mu_0$$

$$\mu_2 = (1 + \chi_{m2})\mu_0 \approx \mu_0$$

Then

$$\beta = \frac{\mu_1(c/n_1)}{\mu_2(c/n_2)} \approx \frac{n_2}{n_1} \approx \frac{v_1}{v_2}$$

$$\tilde{E}_{0-tra} = \frac{2}{1 - \left(\frac{v_1}{v_2}\right)} \tilde{E}_{0-ref}$$



$$\tilde{E}_{0-tra} = \frac{2v_2}{v_2 - v_1} \tilde{E}_{0-ref}$$

$$\tilde{E}_{0-tra} = \frac{2}{1 + \beta} \tilde{E}_{0-inc}$$



$$\tilde{E}_{0-tra} = \frac{2v_2}{v_2 + v_1} \tilde{E}_{0-inc}$$

$$\tilde{E}_{0-ref} = \frac{1 - \beta}{1 + \beta} \tilde{E}_{0-inc}$$



$$\tilde{E}_{0-ref} = \frac{1 - \left(\frac{v_1}{v_2}\right)}{1 + \left(\frac{v_1}{v_2}\right)} \tilde{E}_{0-inc}$$



$$\tilde{E}_{0-ref} = \frac{v_2 - v_1}{v_2 + v_1} \tilde{E}_{0-inc}$$

We can alternatively express these relations in terms of the indices of refraction n_1 & n_2 .

$$\tilde{E}_{0-ref} = \frac{n_1 - n_2}{n_1 + n_2} \tilde{E}_{0-inc}$$

$$\tilde{E}_{0-tra} = \frac{2n_2}{n_2 - n_1} \tilde{E}_{0-ref}$$

$$\tilde{E}_{0-tra} = \frac{2n_1}{n_1 + n_2} \tilde{E}_{0-inc}$$

What fraction of the incident EM wave energy is reflected back from the interface @ $z = 0$?

What fraction of the incident EM wave energy is transmitted through the interface @ $z = 0$?

$$\mathbf{I}(\mathbf{r}, t) = \left(\frac{c}{n}\right) \epsilon E_{0-\text{rms}}^2$$

Watts/m²

$$\mathbf{I}_{\text{rms}}(\mathbf{r}, t) = \frac{1}{2} v \epsilon E_{0-\text{rms}}^2 = \frac{1}{2} \left(\frac{c}{n}\right) \epsilon E_{0-\text{rms}}^2$$

Rms-Watts/m²

Now the ratio of the reflected intensity to the incident intensity is

$$R \equiv \frac{I_R}{I_I} = \left(\frac{\tilde{E}_{0-ref}}{\tilde{E}_{0-inc}} \right)^2$$

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

Whereas the ratio of the transmitted intensity to the incident intensity is

$$T \equiv \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} \left(\frac{\tilde{E}_{0-tra}}{\tilde{E}_{0-inc}} \right)^2$$

$$v_1 = \frac{c}{n_1}$$
$$v_2 = \frac{c}{n_2}$$

$$n_1 = \sqrt{\frac{\varepsilon_1}{\varepsilon_0}}$$

$$n_2 = \sqrt{\frac{\varepsilon_2}{\varepsilon_0}}$$

$$\tilde{E}_{0-tra} = \frac{2n_1}{n_1 + n_2} \tilde{E}_{0-inc}$$

$$T \equiv \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} \left(\frac{\tilde{E}_{0-tra}}{\tilde{E}_{0-inc}} \right)^2$$

$$T = \frac{4n_1 n_2}{(n_1 + n_2)^2}$$

Notice That

$$R + T = 1$$

EM energy is conserved at the interface/boundary between two linear/homogeneous/isotropic media in this process !!!

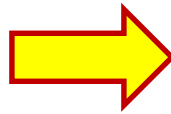
Example: A monochromatic plane EM wave is incident on an air-glass interface (@ $z = 0$) at normal incidence:

Given; Indices of refraction for air and glass ($n_1 = n_{\text{air}} = 1.0$ & $n_2 = n_{\text{glass}} = 1.5$)

Solution:

Reflection Coefficient

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$



$$R = \left(\frac{1 - 1.5}{1 + 1.5} \right)^2$$



$$R = \left(\frac{-0.5}{2.5} \right)^2$$

$$R = 0.04$$

Transmission Coefficient

$$T = \frac{4n_1n_2}{(n_1 + n_2)^2}$$

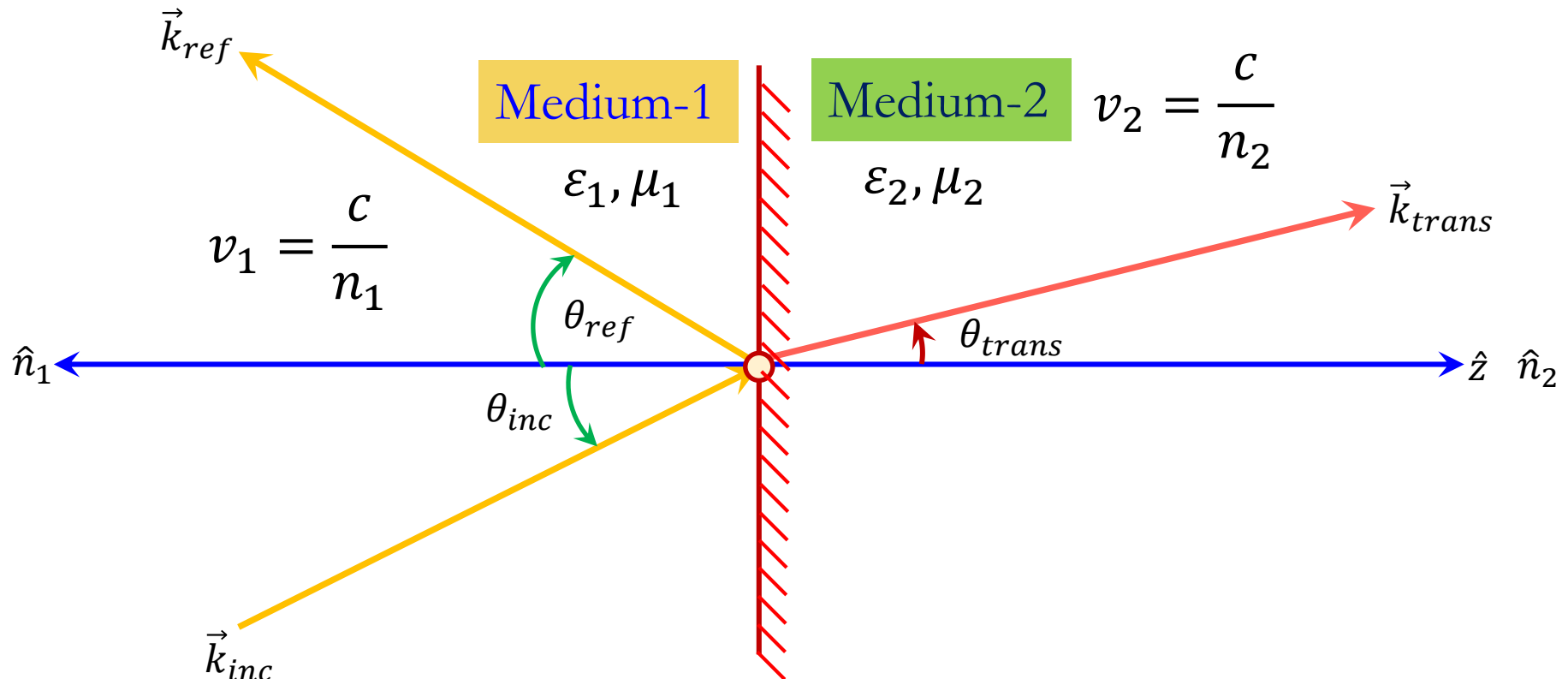
$$T = \frac{4 \times 1 \times 1.5}{(1 + 1.5)^2}$$

$$T = 0.96$$

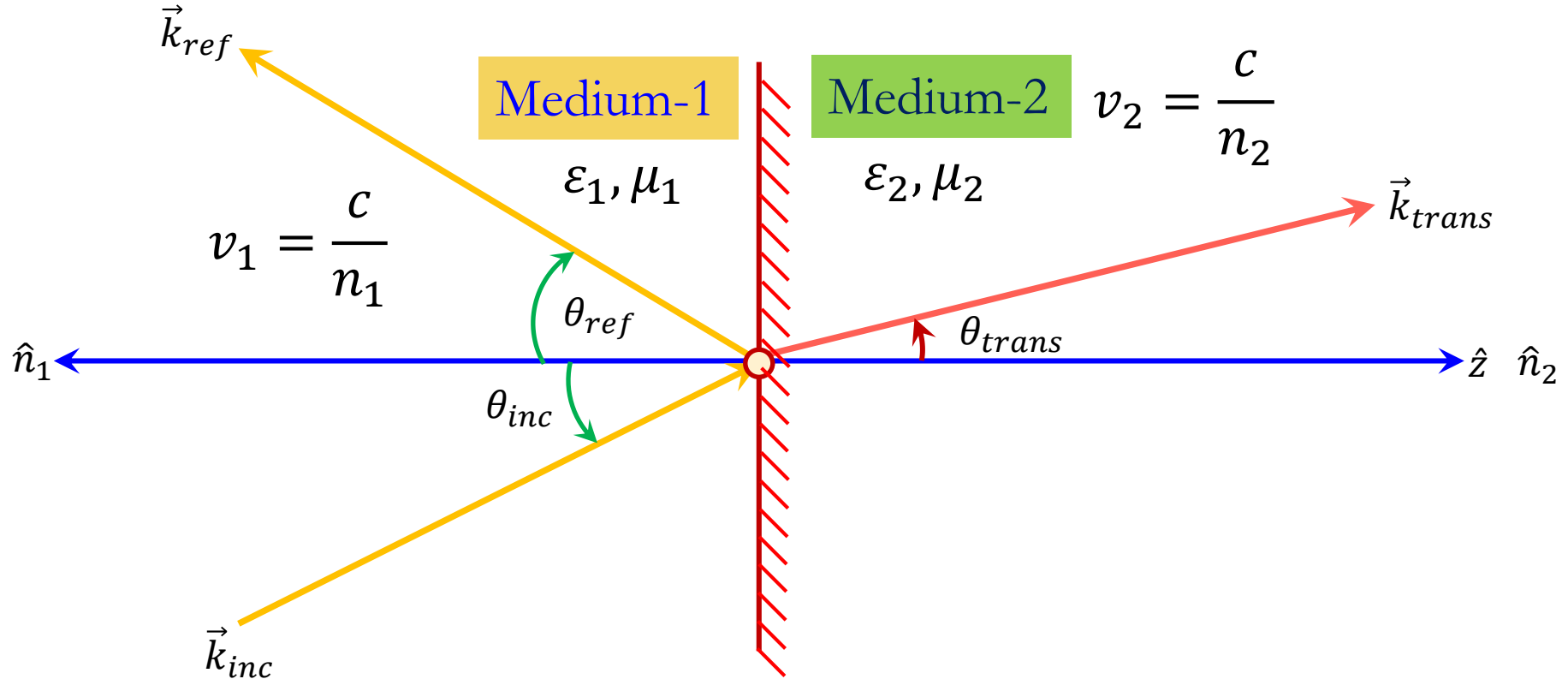
Hence

$$R + T = 0.04 + 0.96 = 1$$

Reflection & Transmission of Monochromatic Plane *EM* Waves at Oblique Incidence at a Boundary Between Two Linear/Homogeneous/Isotropic Media



A monochromatic EM plane wave is incident at an **oblique** angle θ_{inc} on a boundary between two linear/homogeneous/isotropic media. A portion of this EM wave is reflected at angle θ_{ref} , a portion of this EM wave is transmitted, at angle trans θ_{tra} . The three angles are defined with respect to the unit **normal** to the interface $\hat{n}_1, \hat{n}_2$ as shown in the figure below:



The incident EM wave is:

$$\vec{\tilde{E}}_{inc} = \tilde{E}_{0-inc} e^{i(\vec{k}_{inc} \cdot \vec{r} - \omega t)}$$

$$\vec{\tilde{B}}_{inc} = \frac{1}{v_1} \hat{k}_{inc} \times \vec{\tilde{E}}_{inc}$$

The reflected *EM* wave is:

$$\vec{\tilde{E}}_{ref} = \tilde{E}_{0-ref} e^{i(\vec{k}_{ref} \cdot \vec{r} - \omega t)}$$

$$\vec{\tilde{B}}_{ref} = \frac{1}{v_1} \hat{k}_{ref} \times \vec{\tilde{E}}_{ref}$$

The transmitted EM wave is

$$\vec{\tilde{E}}_{tra} = \tilde{E}_{0-tra} e^{i(\vec{k}_{tra} \cdot \vec{r} - \omega t)}$$

$$\vec{\tilde{B}}_{tra} = \frac{1}{v_2} \hat{k}_{tra} \times \vec{\tilde{E}}_{tra}$$

Note that all three EM waves have the same frequency, $f = \frac{\omega}{2\pi}$

This is due to the fact that at the microscopic level, the energy of real photon does *not* change in a medium, *i.e.* $E_{\gamma}^{vac} = E_{\gamma}^{med} = E_{\gamma}$, and since $E_{\gamma} = hf_{\gamma}$ for real photons, then $hf_{\gamma}^{vac} = hf_{\gamma}^{med} = hf_{\gamma}$. Thus, the frequency of a real photon does not change in a medium, *i.e.* $f_{\gamma}^{vac} = f_{\gamma}^{med} = f_{\gamma}$

An experimental fact: colors of objects do not change when placed & viewed e.g. underwater

However, the momentum of a real photon *does* change in a medium! This is because the momentum of the real photon in a medium depends on index of refraction of that medium n_{med} via the relation $p_{\gamma}^{\text{med}} = n_{\text{med}} p_{\gamma}^{\text{vac}}$. Thus the photon momentum depends {inversely} on the speed of propagation in the medium!

From the De-Broglie relation between momentum and wavelength of the real photon $p_{\gamma} = \frac{h}{\lambda_{\gamma}}$ we see that $p_{\gamma}^{\text{med}} = n_{\text{med}} p_{\gamma}^{\text{vac}} = n_{\text{med}} \left(\frac{h}{\lambda_{\gamma}^{\text{vac}}} \right) = \frac{h}{\lambda_{\gamma}^{\text{med}}}$ and hence $\lambda_{\gamma}^{\text{med}} = \frac{\lambda_{\gamma}^{\text{vac}}}{n_{\text{med}}}$

Thus, for *macroscopic* EM waves propagating in the two linear/homogeneous/isotropic media (1) and (2), we have $f_1 = f_2 = f$, and since $\omega = 2\pi f$ then $\omega_1 = \omega_2 = \omega$

$$\omega = kv$$

$$\omega_1 = \omega_2 = \omega \quad \Rightarrow \quad k_1 v_1 = k_2 v_2$$

$$\omega = k_{inc} v_1 = k_{ref} v_1 = k_{tra} v_2$$

$$k_{inc} = |\vec{k}_{inc}| = \frac{2\pi}{\lambda_1}$$

$$k_{ref} = |\vec{k}_{ref}| = \frac{2\pi}{\lambda_1}$$

$$k_{tra} = |\vec{k}_{tra}| = \frac{2\pi}{\lambda_2}$$

$$k_{inc}v_1 = k_{ref}v_1 = k_{tra}v_2 = \omega$$

$$k_{inc} = k_{ref} = k_{tra} \frac{v_2}{v_1} = k_{tra} \frac{n_1}{n_2}$$

The total (*i.e.* combined) EM fields in **medium-1**:

$$\vec{\tilde{E}}_{Tot-1} = \vec{\tilde{E}}_{inc} + \vec{\tilde{E}}_{ref}$$

$$\vec{\tilde{B}}_{Tot-1} = \vec{\tilde{B}}_{inc} + \vec{\tilde{B}}_{ref}$$

$$\vec{\tilde{E}}_{Tot-2} = \vec{\tilde{E}}_{tra}$$

$$\vec{\tilde{B}}_{Tot-2} = \vec{\tilde{B}}_{tra}$$

Now applying the boundary condition (1) and (4) at $z=0$ (i.e. xy plane)

These four boundary conditions generically are of the form:

$$(\quad)e^{i(\vec{k}_{inc}\cdot\vec{r}-\omega t)} + (\quad)e^{i(\vec{k}_{ref}\cdot\vec{r}-\omega t)} = (\quad)e^{i(\vec{k}_{tra}\cdot\vec{r}-\omega t)} \quad \text{at } z=0$$

$$(\quad)e^{i(\vec{k}_{inc}\cdot\vec{r})} + (\quad)e^{i(\vec{k}_{ref}\cdot\vec{r})} = (\quad)e^{i(\vec{k}_{tra}\cdot\vec{r})}$$

For $z = 0$ (i.e. on the interface in the x-y plane) we must have:

$$\vec{k}_{inc} \cdot \vec{r} = \vec{k}_{ref} \cdot \vec{r} = \vec{k}_{tra} \cdot \vec{r}$$

More explicitly:

$$k_{inc_x}x + k_{inc_y}y + \underbrace{k_{inc_z}z}_{z=0} = k_{ref_x}x + k_{ref_y}y + \underbrace{k_{ref_z}z}_{z=0} = k_{tra_x}x + k_{tra_y}y + \underbrace{k_{tra_z}z}_{z=0}$$

$$k_{inc_x}x + k_{inc_y}y = k_{ref_x}x + k_{ref_y}y = k_{tra_x}x + k_{tra_y}y$$

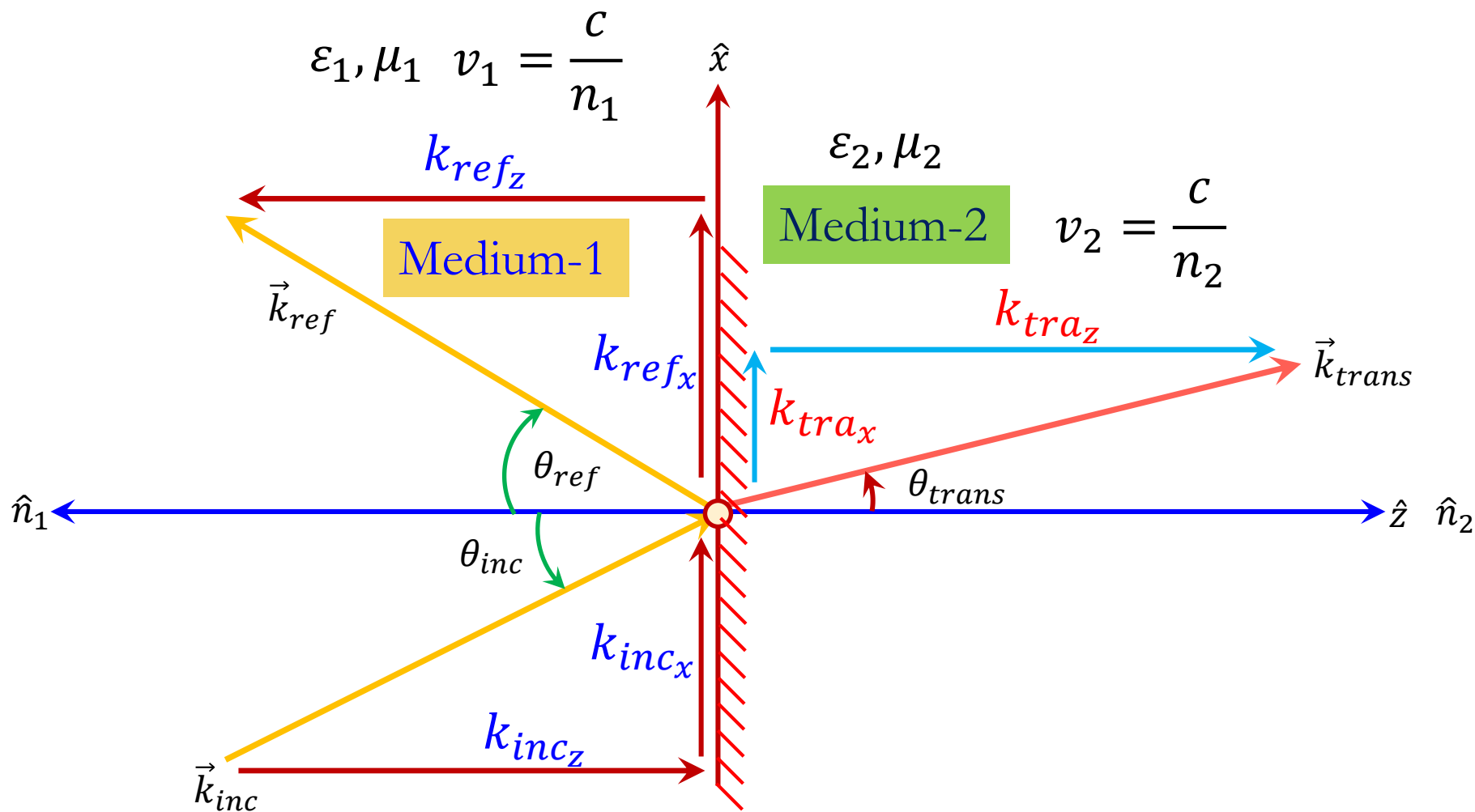
This eq. hold only if the components are separately equal, for if $x=0$, we get

$$k_{inc_y}y = k_{ref_y}y = k_{tra_y}y$$

While for $y = 0$, we get

$$k_{inc_x}x = k_{ref_x}x = k_{tra_x}x$$

Since this problem has ***rotational invariance*** (*i.e.* rotational ***symmetry***) about the $\hat{z}$ axis, without any loss of generality we can *e.g.* choose $\vec{k}_{inc}$ to lie entirely within the x - z plane, as shown in the figure of next slide.



The x-z Plane of Incidence

Then: $k_{inc_y}y = k_{ref_y}y = k_{tra_y}y = 0$

And Thus

$$k_{inc_x} = k_{ref_x} = k_{tra_x}$$

i.e. the transverse components of $\vec{k}_{inc}$, $\vec{k}_{ref}$, $\vec{k}_{tra}$, are all equal and point in the {same} $+\hat{x}$ direction.

The First Law of Geometrical Optics (Law of Reflection)

All wave-vectors k lie in a common plane

The above result tells us that the three wave vectors $\vec{k}_{inc}$, $\vec{k}_{ref}$ and $\vec{k}_{tra}$ all lie in a plane.

First Law: The incident, reflected, and transmitted wave vectors form a plane (called the plane of incidence), which also includes the normal to the surface.

The Second Law of Geometrical Optics (Law of Reflection)

From the figure, we see that:

$$k_{inc_x} = k_{inc} \sin \theta_{inc} \implies k_{ref_x} = k_{ref} \sin \theta_{ref}$$

$\parallel$

$$k_{tra_x} = k_{tra} \sin \theta_{tra}$$

But

$$k_{inc} = k_{ref}$$



$$k_{inc} \sin \theta_{inc} = k_{ref} \sin \theta_{ref}$$

$$\sin \theta_{inc} = \sin \theta_{ref}$$

$$\theta_{inc} = \theta_{ref}$$

This is the law of reflection: The angle of incidence is equal to the angle of reflection.

The third Law of Geometrical Optics (Law of Refraction)

From the figure, we see that:

$$k_{inc_x} = k_{inc} \sin \theta_{inc}$$

In medium (1):

$$k_{inc} = k_1 = \frac{\omega}{v_1} = \frac{n_1 \omega}{c} = n_1 k_0$$

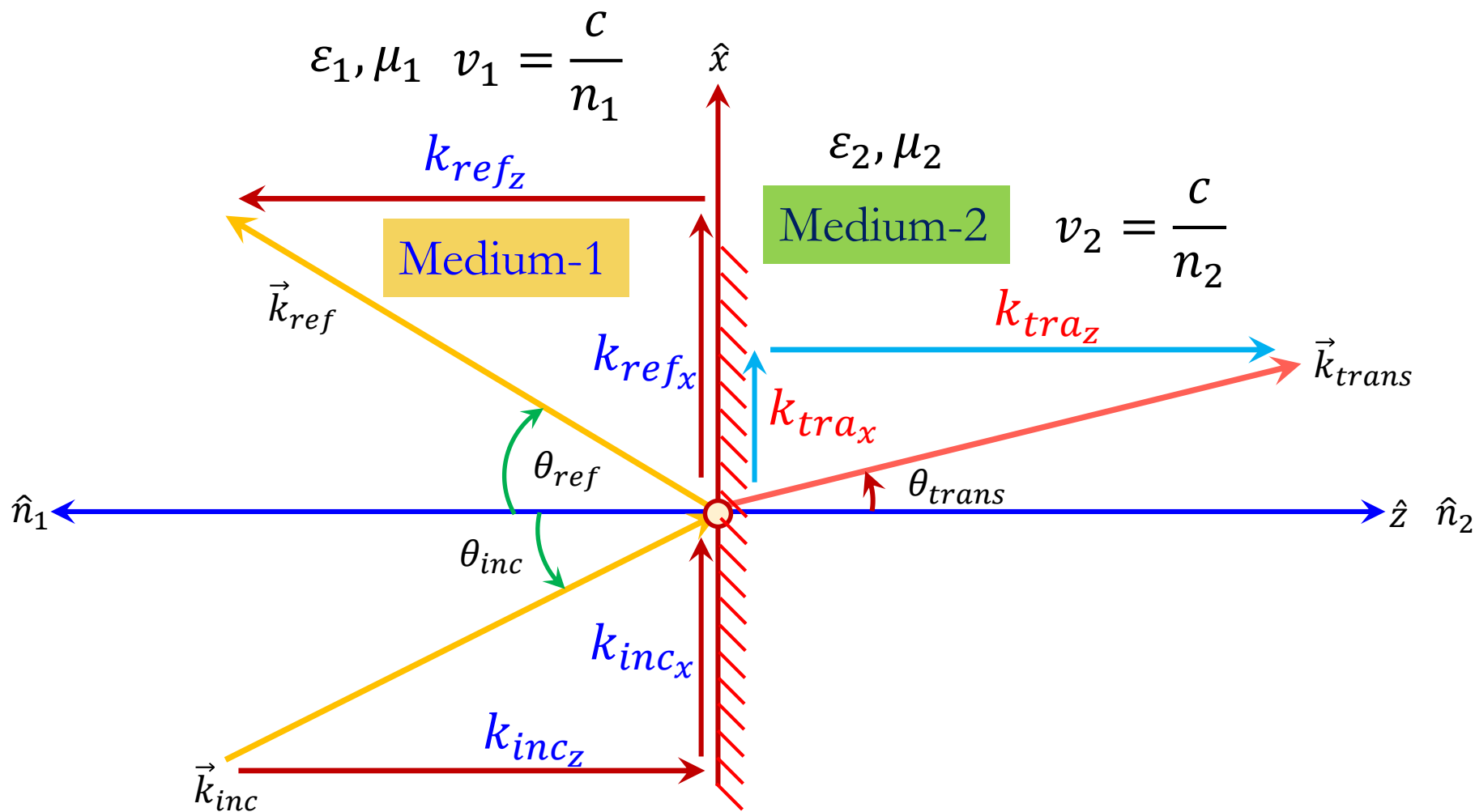
$$k_0 = \text{vacuum wave number} = \frac{2\pi}{\lambda_0}$$

$$\lambda_0 = \text{vacuum wavelength}$$

In medium (2):

$$k_{tra} = k_2 = \frac{\omega}{v_2} = \frac{n_2 \omega}{c} = n_2 k_0$$

$$k_{inc} \sin \theta_{inc} = k_{tra} \sin \theta_{tra}$$



The x-z Plane of Incidence

$$k_{inc} = n_1 k_0$$

$$k_{tra} = n_2 k_0$$

$$n_1 k_0 \sin \theta_{inc} = n_2 k_0 \sin \theta_{tra}$$

$$n_1 k_0 \sin \theta_{inc} = n_2 k_0 \sin \theta_{tra}$$

$$\frac{\sin \theta_{tra}}{\sin \theta_{inc}} = \frac{n_1}{n_2}$$

This is the law of reflection-Snell's Law

Now that we have taken care of the exponential factors, they cancel, The boundary conditions yields the results:

Boundary-Condition 1:

$$\varepsilon_1 E_{1-Tot}^{\perp} = \varepsilon_2 E_{2-Tot}^{\perp}$$

$$\varepsilon_1 \left(\vec{\tilde{E}}_{inc} + \vec{\tilde{E}}_{ref} \right)_z = \varepsilon_2 \left(\vec{\tilde{E}}_{tra} \right)_z$$

Boundary-Condition 2:

$$E_{1-Tot}^{\parallel} = E_{2-Tot}^{\parallel}$$

$$\left(\vec{\tilde{E}}_{inc} + \vec{\tilde{E}}_{ref} \right)_{xy} = \left(\vec{\tilde{E}}_{tra} \right)_{xy}$$

Boundary-Condition 3:

$$B_{1-Tot}^{\perp} = B_{2-Tot}^{\perp}$$

$$\left(\vec{\tilde{B}}_{inc} + \vec{\tilde{B}}_{ref}\right)_z = \left(\vec{\tilde{B}}_{tra}\right)_z$$

Boundary-Condition 4:

$$\frac{1}{\mu_1} B_{1-Tot}^{\parallel} = \frac{1}{\mu_2} B_{2-Tot}^{\parallel}$$

$$\frac{1}{\mu_1} \left(\vec{\tilde{B}}_{inc} + \vec{\tilde{B}}_{ref}\right)_{xy} = \frac{1}{\mu_2} \left(\vec{\tilde{B}}_{tra}\right)_{xy}$$

We also have the relation

$$\vec{B}_0 = \frac{1}{v} \hat{k} \times \vec{E}_0$$

For a *EM* plane wave incident on a boundary between two linear/homogeneous/isotropic media at an ***oblique*** angle of incidence, there are ***three*** possible **polarization** cases to consider:

Case-I: $\vec{E}_{inc} \perp$ plane of incidence – known as Transverse Electric (TE) Polarization

$$\vec{B}_{inc} \parallel \text{plane of incidence}$$

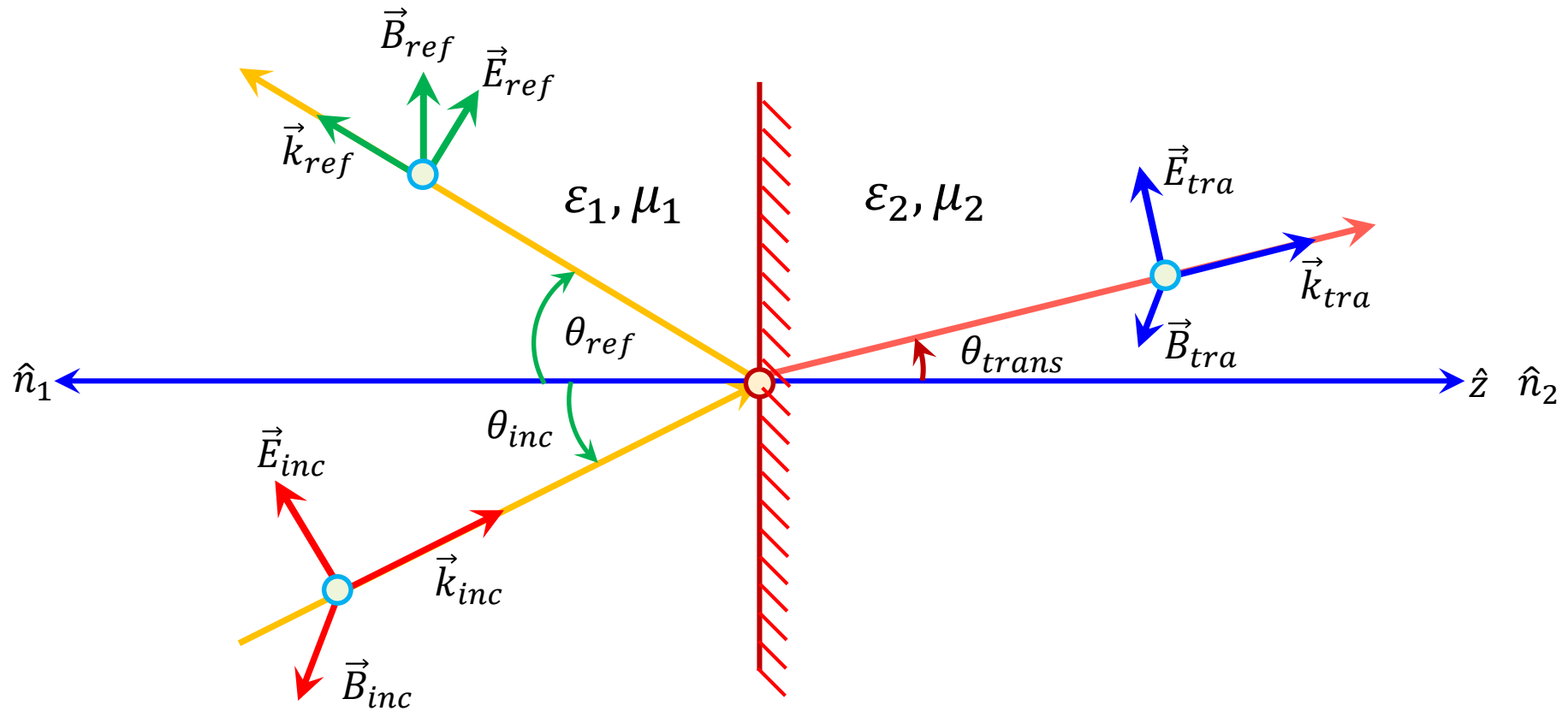
Case-II: $\vec{E}_{inc} \parallel$ plane of incidence, known as Transverse Magnetic (TM) Polarization

$$\vec{B}_{inc} \perp \text{plane of incidence}$$

Case-III: The most general case: $\vec{E}_{inc}$ is neither $\parallel$ nor $\perp$ to the plane of incidence. It's a linear combination of case-I and case-II.

Here we will solve only case-II and others were leave as an take home exercise.

Case-II: Transverse Magnetic (TM) Polarization



Medium-1 $v_1 = \frac{c}{n_1}$

Medium-2 $v_2 = \frac{c}{n_2}$

For Case-II: The Boundary Conditions becomes

Boundary-Condition 1:

$$\varepsilon_1 \left(\vec{\tilde{E}}_{inc} + \vec{\tilde{E}}_{ref} \right)_z = \varepsilon_2 \left(\vec{\tilde{E}}_{tra} \right)_z$$

$$\varepsilon_1 \left(-\tilde{E}_{o-inc} \sin \theta_{inc} + \tilde{E}_{o-ref} \sin \theta_{ref} \right) = \varepsilon_2 \left(-\tilde{E}_{o-tra} \sin \theta_{tra} \right)$$

$$\theta_{inc} = \theta_{ref}$$

Law of reflection

$$\varepsilon_1 \left(-\tilde{E}_{o-inc} \sin \theta_{inc} + \tilde{E}_{o-ref} \sin \theta_{inc} \right) = \varepsilon_2 \left(-\tilde{E}_{o-tra} \sin \theta_{tra} \right)$$

$$\left(-\tilde{E}_{o-inc} + \tilde{E}_{o-ref} \right) = -\frac{\varepsilon_2}{\varepsilon_1} \left(\frac{\sin \theta_{tra}}{\sin \theta_{inc}} \right) \tilde{E}_{o-tra}$$

$$\frac{\sin \theta_{tra}}{\sin \theta_{inc}} = \frac{n_1}{n_2}$$

Snell's law

$$\frac{n_1}{n_2} = \frac{v_2}{v_1}$$

$$v = \frac{1}{\sqrt{\epsilon\mu}}$$

$$(\tilde{E}_{o-inc} - \tilde{E}_{o-ref}) = \frac{\epsilon_2}{\epsilon_1} \left(\frac{n_1}{n_2} \right) \tilde{E}_{o-tra}$$

$$(\tilde{E}_{o-inc} - \tilde{E}_{o-ref}) = \beta \tilde{E}_{o-tra}$$

$$\beta = \frac{v_1 \mu_1}{v_2 \mu_2} = \frac{n_2 \mu_1}{n_1 \mu_2}$$

Boundary-Condition 2:

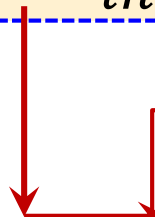
$$\tilde{E}_{o-inc} \cos \theta_{inc} + \tilde{E}_{o-ref} \cos \theta_{ref} = \tilde{E}_{o-tra} \cos \theta_{tra}$$

$$\theta_{inc} = \theta_{ref}$$

Law of reflection

$$\tilde{E}_{o-inc} \cos \theta_{inc} + \tilde{E}_{o-ref} \cos \theta_{inc} = \tilde{E}_{o-tra} \cos \theta_{tra}$$

$$\tilde{E}_{o-inc} + \tilde{E}_{o-ref} = \tilde{E}_{o-tra} \frac{\cos \theta_{tra}}{\cos \theta_{inc}}$$


$$\alpha = \frac{\cos \theta_{tra}}{\cos \theta_{inc}}$$

$$\tilde{E}_{o-inc} + \tilde{E}_{o-ref} = \alpha \tilde{E}_{o-tra}$$

Boundary Condition-3 Contributes Nothing since B does not have z-component.

Boundary-Condition 4:

$$\frac{1}{\mu_1} \left(\vec{\tilde{B}}_{inc} + \vec{\tilde{B}}_{ref} \right)_{xy} = \frac{1}{\mu_2} \left(\vec{\tilde{B}}_{tra} \right)_{xy}$$

Here B is in $\hat{y}$ -direction

$$\frac{1}{\mu_1} \left(\vec{\tilde{B}}_{inc} + \vec{\tilde{B}}_{ref} \right)_y = \frac{1}{\mu_2} \left(\vec{\tilde{B}}_{tra} \right)_y$$

$$\vec{\tilde{B}}_0 = \frac{1}{v} \hat{k} \times \vec{\tilde{E}}_0$$

$$\frac{1}{\mu_1 v_1} \left(\vec{\tilde{E}}_{o-inc} - \vec{\tilde{E}}_{o-ref} \right) = \frac{1}{\mu_2 v_2} \left(\vec{\tilde{E}}_{o-tra} \right)$$

$$(\tilde{E}_{o-inc} - \tilde{E}_{o-ref}) = \frac{\mu_1 v_1}{\mu_2 v_2} (\tilde{E}_{o-tra})$$

$$(\tilde{E}_{o-inc} - \tilde{E}_{o-ref}) = \beta (\tilde{E}_{o-tra})$$

This result is same as from Boundary condition (1)

Now solving the equations:

$$(\tilde{E}_{o-inc} - \tilde{E}_{o-ref}) = \beta \tilde{E}_{o-tra}$$

$$\tilde{E}_{o-inc} + \tilde{E}_{o-ref} = \alpha \tilde{E}_{o-tra}$$

$$\tilde{E}_{o-ref} = \left(\frac{\alpha - \beta}{\alpha + \beta} \right) \tilde{E}_{o-inc}$$

$$\tilde{E}_{o-tra} = \left(\frac{2}{\alpha + \beta} \right) \tilde{E}_{o-inc}$$

Fresnel's Equations

(For the case of polarization in the plane of incidence)

Notice that the transmitted wave is always in phase with the incident one; the reflected wave is either in phase ("right side up"), if $\alpha > \beta$, or 180° out of phase ("upside down"). if $\alpha < \beta$.

The amplitudes of the transmitted and reflected waves depend on the angle of incidence, because ' α ' is a function of θ_{inc} :

$$\alpha = \frac{\sqrt{1 - \sin^2 \theta_{tra}}}{\cos \theta_{inc}}$$

$$\frac{\sin \theta_{tra}}{\sin \theta_{inc}} = \frac{n_1}{n_2}$$

Snell's law

$$\alpha = \frac{\sqrt{1 - (n_1/n_2 \sin \theta_{inc})^2}}{\cos \theta_{inc}}$$

In the case of Normal incidence ($\theta_{inc} = 0$), $\alpha = 1$, and we recover the equations obtained in the slides# 36 i.e Normal incidence case.

$$\tilde{E}_{0-ref} = \frac{n_1 - n_2}{n_1 + n_2} \tilde{E}_{0-inc}$$

$$\tilde{E}_{0-tra} = \frac{2n_2}{v_2 - v_1} \tilde{E}_{0-ref}$$

At grazing incidence ($\theta_{inc} = 90^\circ$), α diverges, and the wave is totally reflected (a fact that is painfully familiar to anyone who has driven at night on a wet road). Interestingly, there is an intermediate angle, θ_B = called Brewster's angle, at which the reflected wave is completely extinguished.

According to the equations

$$\tilde{E}_{o-ref} = \left(\frac{\alpha - \beta}{\alpha + \beta} \right) \tilde{E}_{o-inc}$$

When reflected wave is completely extinguished (i.e. $\tilde{E}_{o-ref} = 0$)

This will happen when $\alpha = \beta$

$$\alpha = \frac{\sqrt{1 - (n_1/n_2 \sin \theta_{inc})^2}}{\cos \theta_{inc}}$$

$$\alpha = \beta$$

$$\theta_{inc} = \theta_B$$

$$\beta^2 \cos^2 \theta_B = 1 - (n_1/n_2 \sin \theta_B)^2$$

$$\beta^2 (1 - \sin^2 \theta_B) = 1 - (n_1/n_2 \sin \theta_B)^2$$

$$\beta^2(1 - \sin^2 \theta_B) = 1 - (n_1/n_2 \sin \theta_B)^2$$

Solving for $\sin^2 \theta_B$

$$\sin^2 \theta_B ((n_1/n_2)^2 - \beta^2) = 1 - \beta^2$$

$$\sin^2 \theta_B = \frac{1 - \beta^2}{((n_1/n_2)^2 - \beta^2)}$$

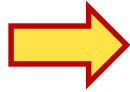
For the typical case $\mu_1 \cong \mu_2$, so $\beta \cong \frac{n_2}{n_1}$,

$$\sin^2 \theta_B = \frac{1 - \beta^2}{((1/\beta)^2 - \beta^2)}$$

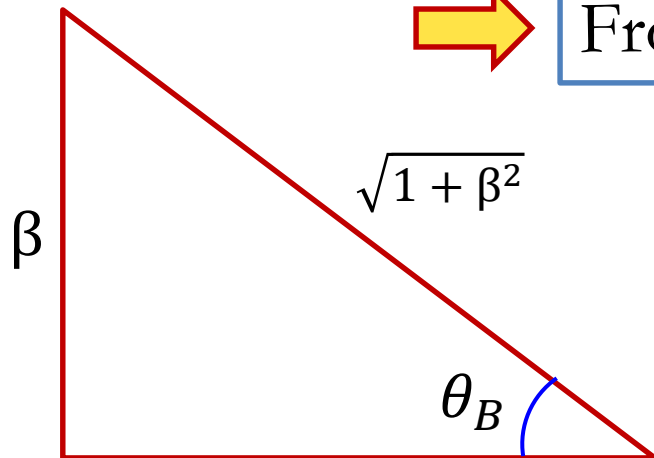
$$\sin^2 \theta_B = \frac{(1 - \beta^2)\beta^2}{((1 - \beta^4))}$$

$$\sin^2 \theta_B = \frac{(1 - \beta^2)\beta^2}{((1 - \beta^2)(1 + \beta^2))}$$

$$\sin^2 \theta_B = \frac{\beta^2}{(1 + \beta^2)}$$



From the Geometry



$$\tan \theta_B = \frac{n_2}{n_1}$$

A very well known
expression

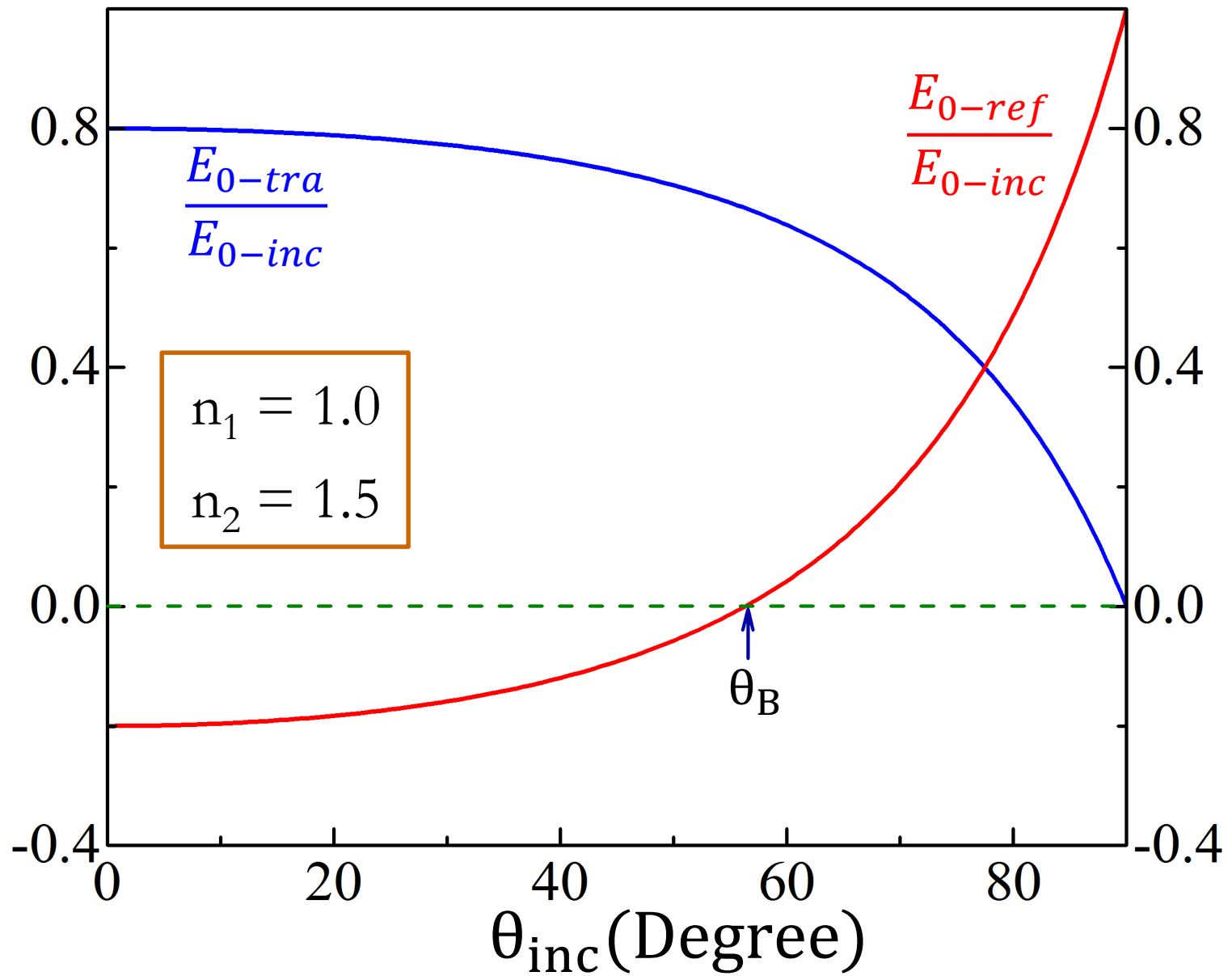


Figure shows a plot of the transmitted and reflected amplitudes as functions of θ_{inc} , for light incident on glass ($n_2 = 1.5$) from air ($n_1 = 1$). (On the graph, a negative number indicates that the wave is 180° out of phase with the incident beam-the amplitude itself is the absolute value.)

The power per unit area striking the interface is $\vec{S} \cdot \hat{z}$. Thus the incident intensity is

$$I_{inc} = \frac{1}{2} \varepsilon_1 v_1 E_{o-inc}^2 \cos \theta_{inc}$$

While the reflected and transmitted intensities are

$$I_{ref} = \frac{1}{2} \varepsilon_1 v_1 E_{o-ref}^2 \cos \theta_{ref} \quad I_{tra} = \frac{1}{2} \varepsilon_2 v_2 E_{o-tra}^2 \cos \theta_{tra}$$

The reflection and transmission coefficients for waves polarized parallel to the plane of incidence are

$$R \cong \frac{I_R}{I_I} = \left(\frac{E_{o-ref}}{E_{o-inc}} \right)^2 = \left(\frac{\alpha - \beta}{\alpha + \beta} \right)^2$$

$$T \cong \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} \left(\frac{E_{o-tra}}{E_{o-inc}} \right)^2 \frac{\cos \theta_{tra}}{\cos \theta_{inc}} = \alpha \beta \left(\frac{2}{\alpha + \beta} \right)^2$$

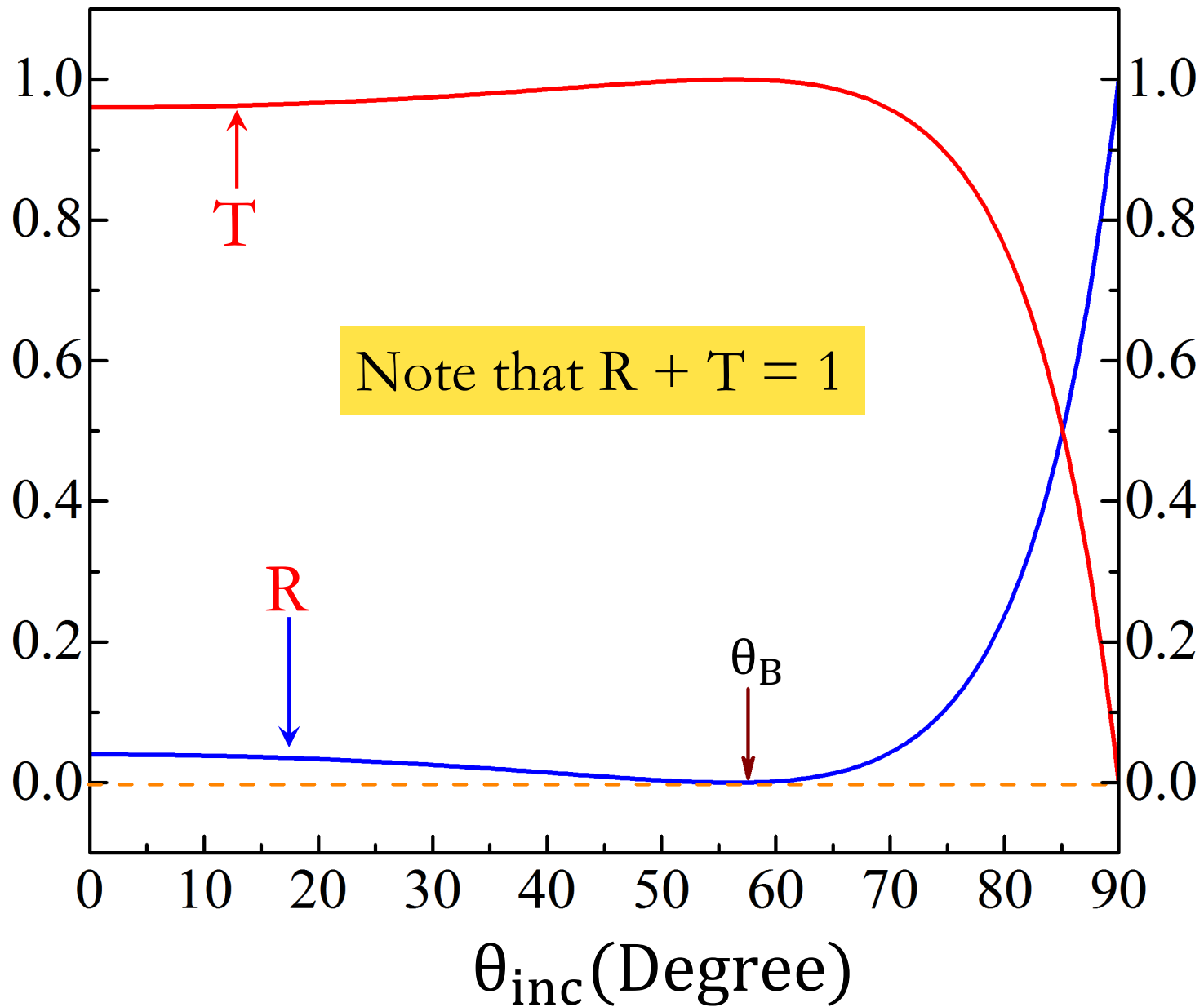


Figure shows a plot of the transmitted (T) and reflected (R) coefficients as functions of θ_{inc} .

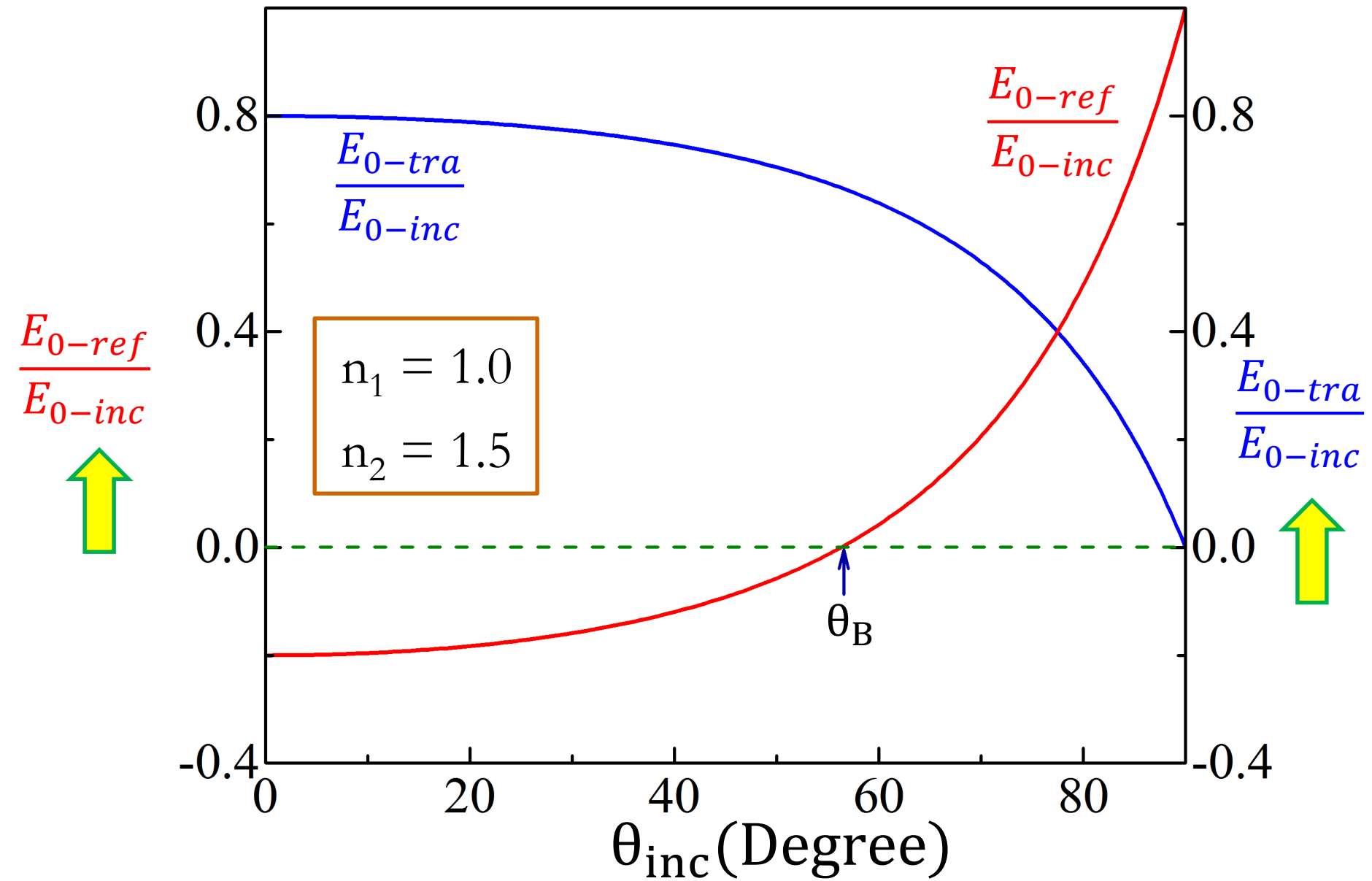
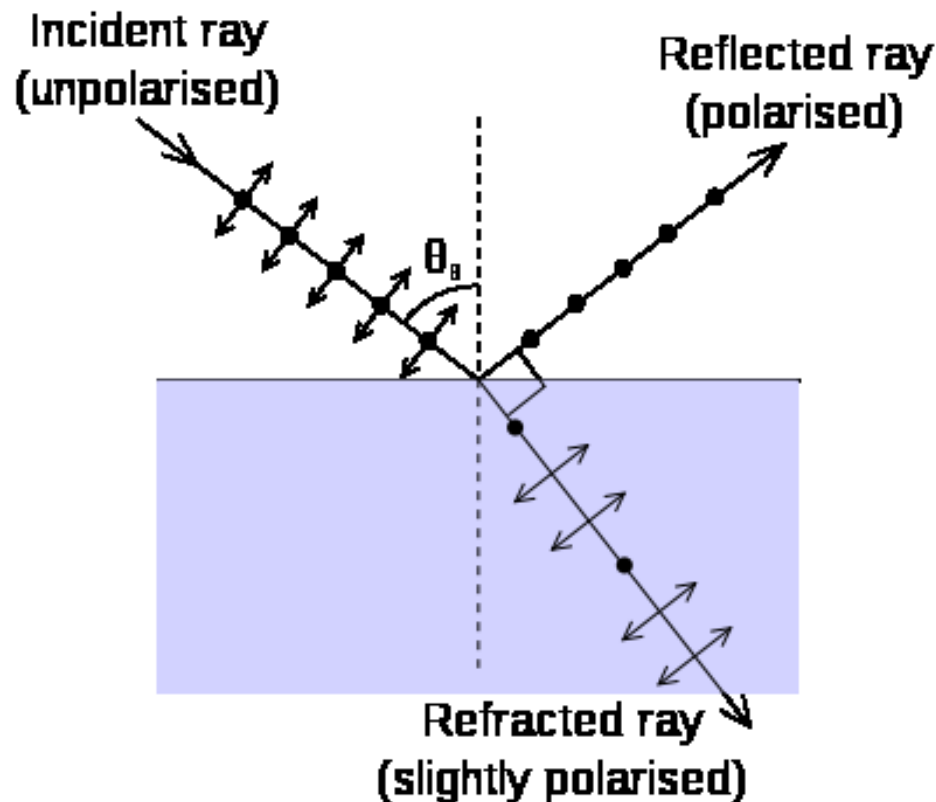


Figure shows a plot of the transmitted and reflected amplitudes as functions of θ_{inc} , for light incident on glass ($n_2 = 1.5$) from air ($n_1 = 1$). (On the graph, a negative number indicates that the wave is 180° out of phase with the incident beam-the amplitude itself is the absolute value.)

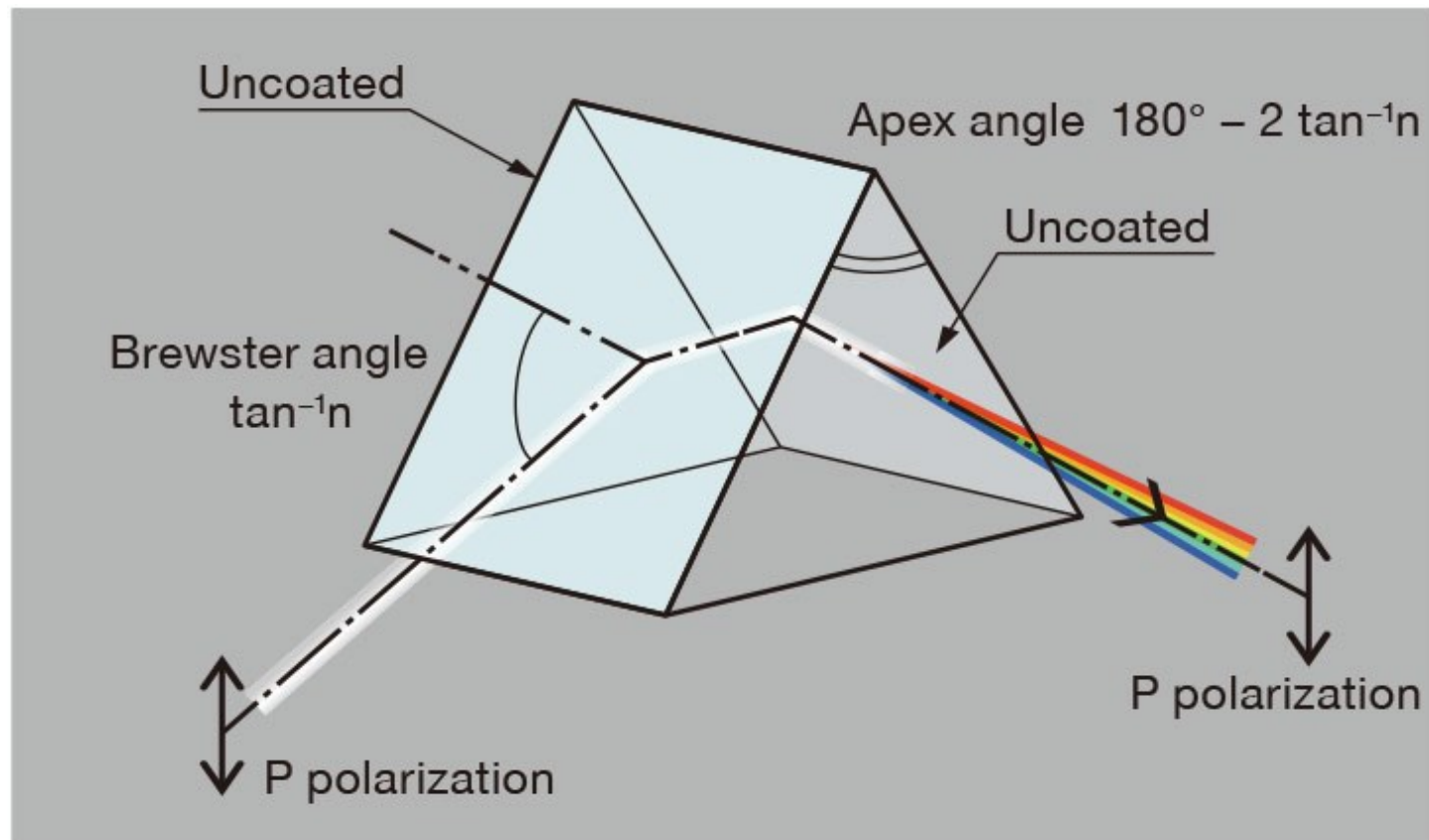
Brewster's law, relationship for light waves stating that the **maximum polarization** (vibration in one plane only) of a ray of light may be achieved by letting the ray fall on a surface of a transparent medium in such a way that the refracted ray makes an angle of 90° with the reflected ray.

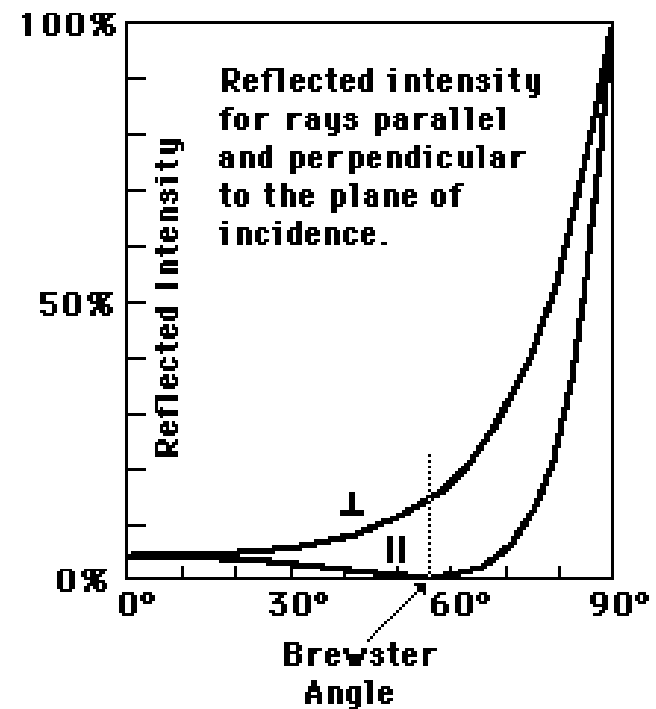
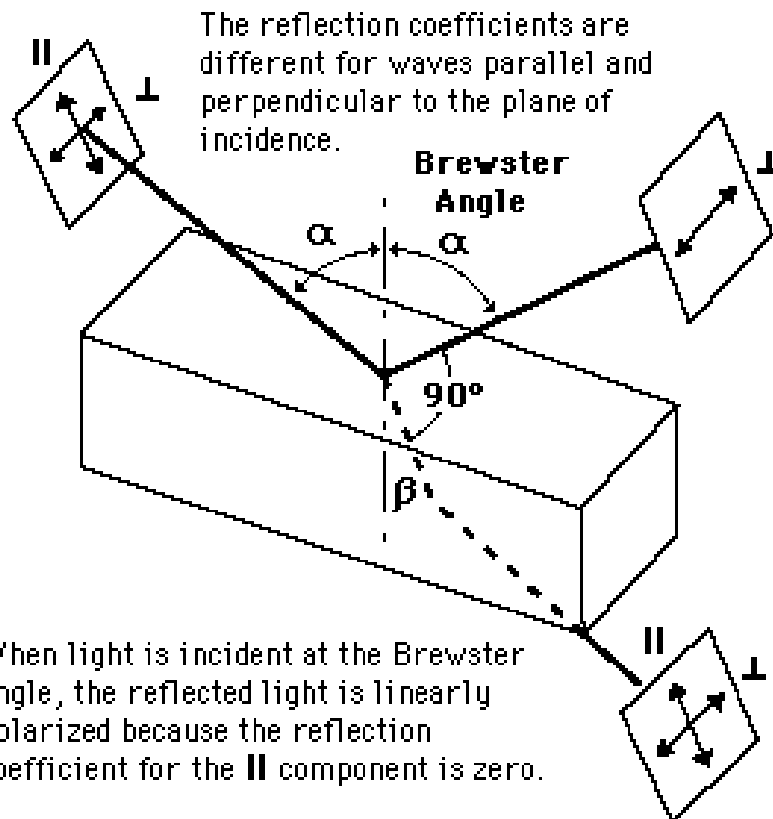
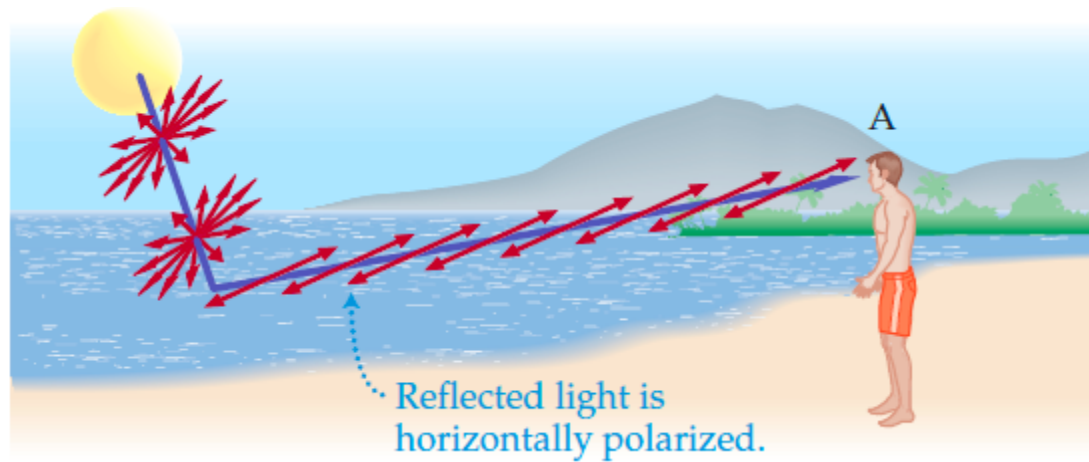
This law is named after a Scottish physicist, **Sir David Brewster**, who first proposed it in 1811.



In Hologram: When recording a hologram, light is typically incident at Brewster's angle. Because the incident light is polarized, it is not back reflected from the transparent back-plane of the holographic film. This avoids unwanted interference effects in the hologram.

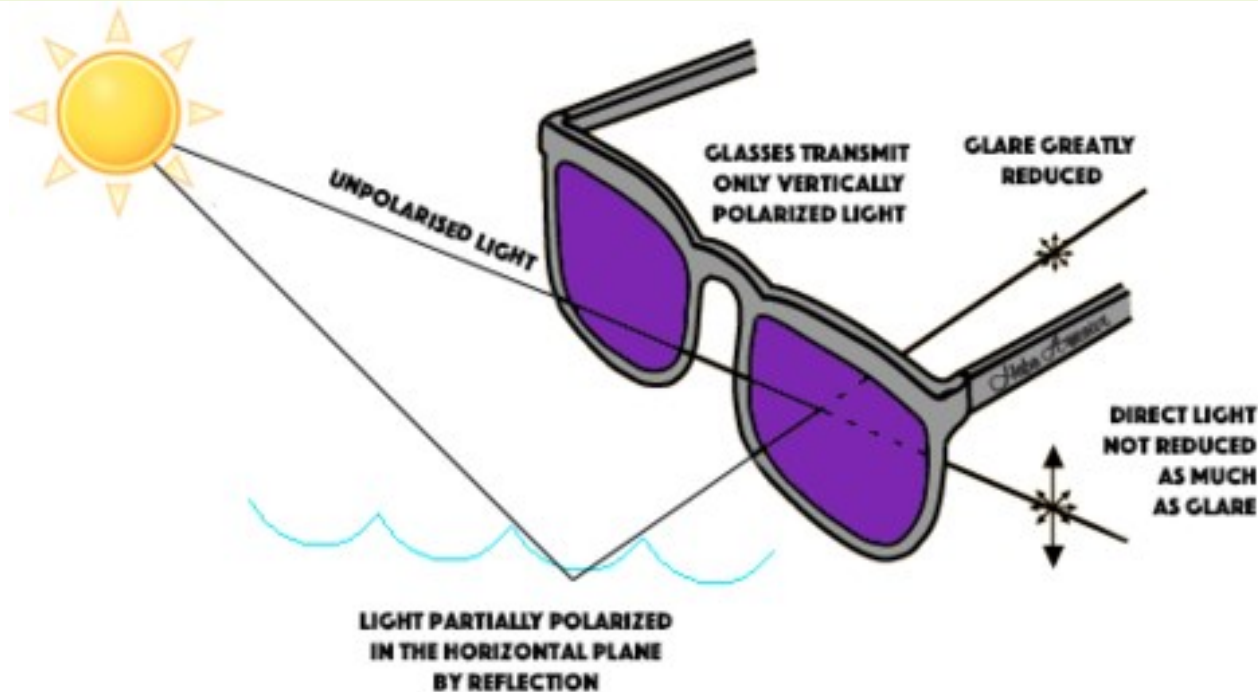
Brewster angle prisms are used in laser physics. The polarized laser light enters the prism at Brewster's angle **without any reflective losses**.





Some Applications of Brewster's Law

Polarized sunglasses use the principle of Brewster's angle to reduce glare from the sun reflecting off horizontal surfaces such as water or road.



When light is reflected from a non-metallic surface, such as a tabletop, snow, or water, it becomes polarized in a plane parallel to the surface. In Polaroid sunglasses, the axes of polarization are vertical. The reflected glare, which is at least partially polarized in the horizontal direction, is reduced by the Polaroid sunglasses.

Find a surface (e.g., tabletop, blackboard, etc.) from which the reflected light produces glare. While viewing the glare through a Polaroid filter, rotate the filter until you no longer see the reflected light. In this position, the filter's axis of polarization is vertical. Place a small piece of masking tape along the edge of the filter and indicate the axis of polarization with an arrow. Rotate the filter to pass the maximum amount of light. Describe the orientation of the axis of polarization now. Now view reflected light from a sheet of metal such as a piece of aluminum foil. Describe what happens this time as you rotate the filter. Compare the reflected light from a metallic surface to that reflected from a non-metallic surface.

Getting Quantitative: Finding Brewster's Angle:

View the glare from a surface through a Polaroid filter held close to one eye. When the axis of polarization of the filter is vertical, you will notice that the reflected light is dim for a variety of viewing angles but completely dark for only one. The angle of reflection that produces completely polarized light is called Brewster's angle and depends on the reflecting surface.

Measure Brewster's angle for one or two different reflecting surfaces. To do this, have your partner extend a string from the filter to the spot on the surface where the light is totally extinguished. Use a protractor to determine the angle formed by a normal (perpendicular line) to the surface and the taut string.

Brewster's angle for water is approximately 53 degrees; for glass, 56 degrees. You may wish to find Brewster's angle for plastic or floor wax by using them as your reflecting surfaces.